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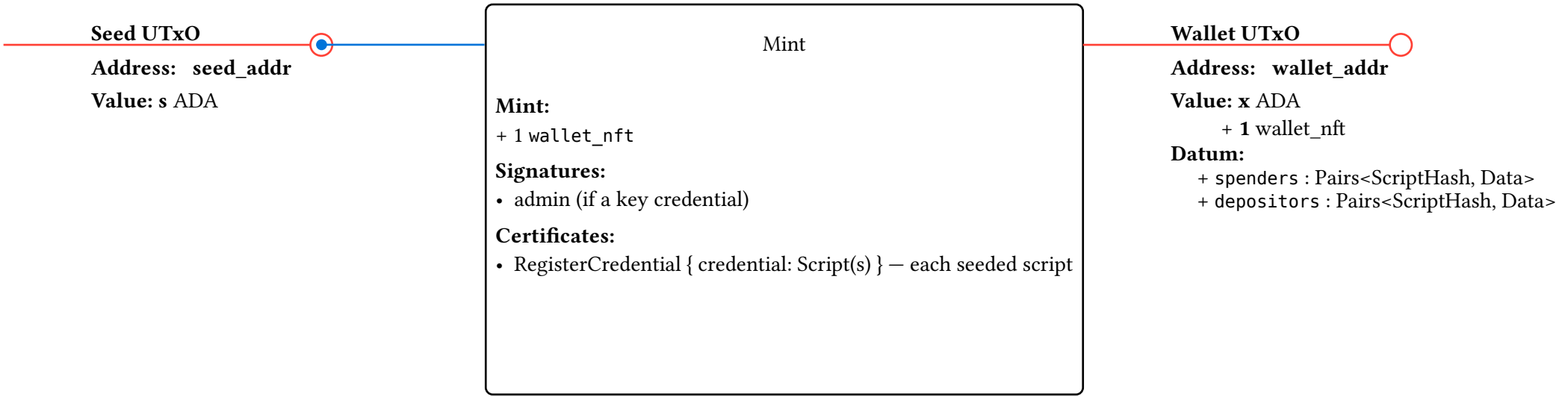
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1 - Mint (create the wallet)

The wallet is created by minting its identifying NFT — one token under the script’s own policy — into a fresh wallet UTxO at the wallet address. The admin must authorize the mint.

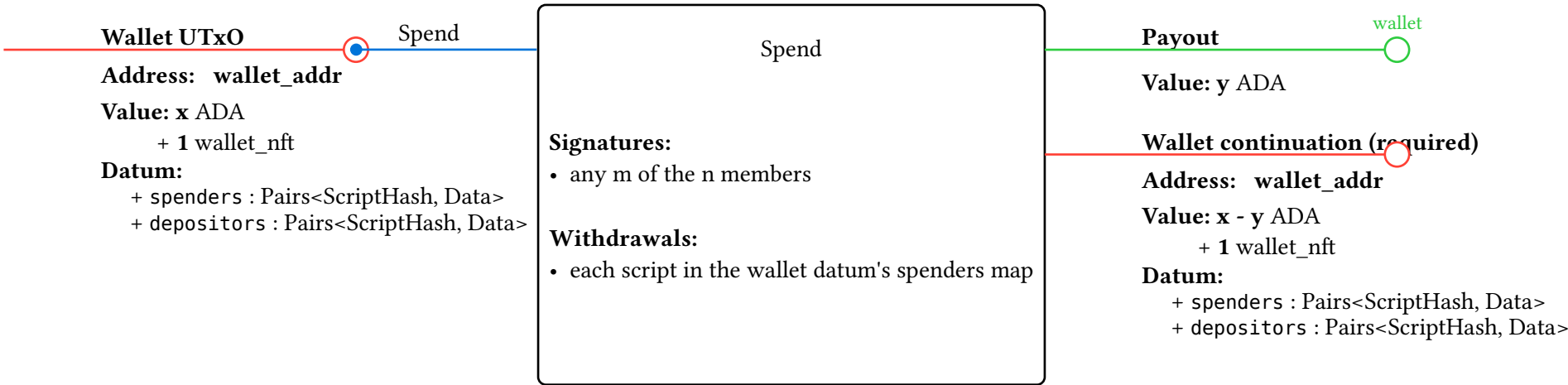


- Notes:**
- wallet_addr: payment is wallet_hash, staking any. wallet_hash = hash(smart_wallet(seed_utxo, settings_ref, wallet_nft_name, admin)) — the multivalidator is parameterized by its seed, configuration reference, NFT name, and the admin credential, so all wallet UTxOs share one address and one NFT policy.
 - one-shot: the mint requires the seed_utxo to be spent, and a UTxO can be spent at most once, so this policy id mints exactly one token ever. The resulting NFT is unforgeable and identifies the wallet.
 - the mint redeemer is Mint { out_ix }, locating the new wallet UTxO.
 - the new wallet UTxO carries the NFT, an inline datum with spenders and depositors maps, and no reference script. Every script in both maps must be registered in this same transaction — a RegisterCredential certificate for its stake credential, which runs the script’s publish handler. The handler reads its own initial Data from this output’s datum and validates it, so a wallet may be created already carrying delegated restrictions or deposit validations, but only ones that self-validated their initial state.
 - the admin credential (a script parameter) must be satisfied: a key admin signs; a script admin authorizes by a withdraw-0 invocation.

Figure 1: Mint (create the wallet)

2 - Spend (base M-of-N + delegated authorizations)

The wallet pays out. The spending script enforces its base M-of-N signature floor and forwards every other check to the withdrawal scripts named in the wallet UTxO's datum; the single spend redeemer asserts their conjunction. The wallet must survive the spend: an NFT-bearing continuation carries depositors forward unchanged.



Notes:

- wallet_addr: payment is wallet_hash, staking any. A wallet UTxO holds its identifying NFT plus ada and arbitrary native assets.
- wallet_hash = hash(smart_wallet(seed_utxo, settings_ref, wallet_nft_name, admin)): the spending script is parameterized by its seed, configuration reference, NFT name, and the admin credential, so all wallet UTxOs share one address. Externalizing the members/threshold config lets it change without redeploying the script.
- validity: the spend requires the wallet UTxO to carry its NFT (1 wallet_nft under wallet_hash) — only NFT-bearing UTxOs are valid smart wallets.
- base M-of-N floor: the wallet's configuration supplies members and threshold, and the spend requires at least threshold of members present in extra_signatories. A member may itself be a script credential, so the floor can be another multisig, a DAO, and so on. This design is agnostic of how that configuration is sourced.
- delegated authorizations: every ScriptHash key in the wallet UTxO's spenders datum field is a withdraw-0 staking script that must run in this transaction; the Data value carries that script's own mutable state. The Spend redeemer asserts the conjunction — the M-of-N floor **and** all delegated checks — so no restriction can be silently omitted. The map is per-UTxO, so different wallet UTxOs may carry different restriction sets.
- the delegated scripts are the extension point: spending limits, time locks, recipient allow-lists, oracle attestations, ... each is an independent withdrawal script observed here, none known to the spending script.
- wallet survival: the transaction must recreate the wallet — an NFT-bearing output at the wallet address carrying the inline datum and a value that is a subset of the input's (a spend only removes value; additions belong to Deposit). Its depositors map must equal the input's and the spenders key set must stay fixed (only UpdatePermissions may change either), while spender Data may change: each spender script enforces its own transition. The NFT can therefore only leave the wallet on Close.
- composability: the spend asserts only its own input, the required spenders, and the surviving wallet with unchanged depositors, a fixed spenders key set, and a value no greater than the input's — never the total input or output counts, nor unrelated UTxOs. Payout outputs are otherwise unconstrained.

Figure 2: Spend (base M-of-N + delegated authorizations)

3 - Update config (per-UTxO)

An authorized admin credential — a parameter of the spending script — rewrites a wallet UTxO’s config (spenders and depositors) in place. Funds are untouched: same address, same value — only the config changes.

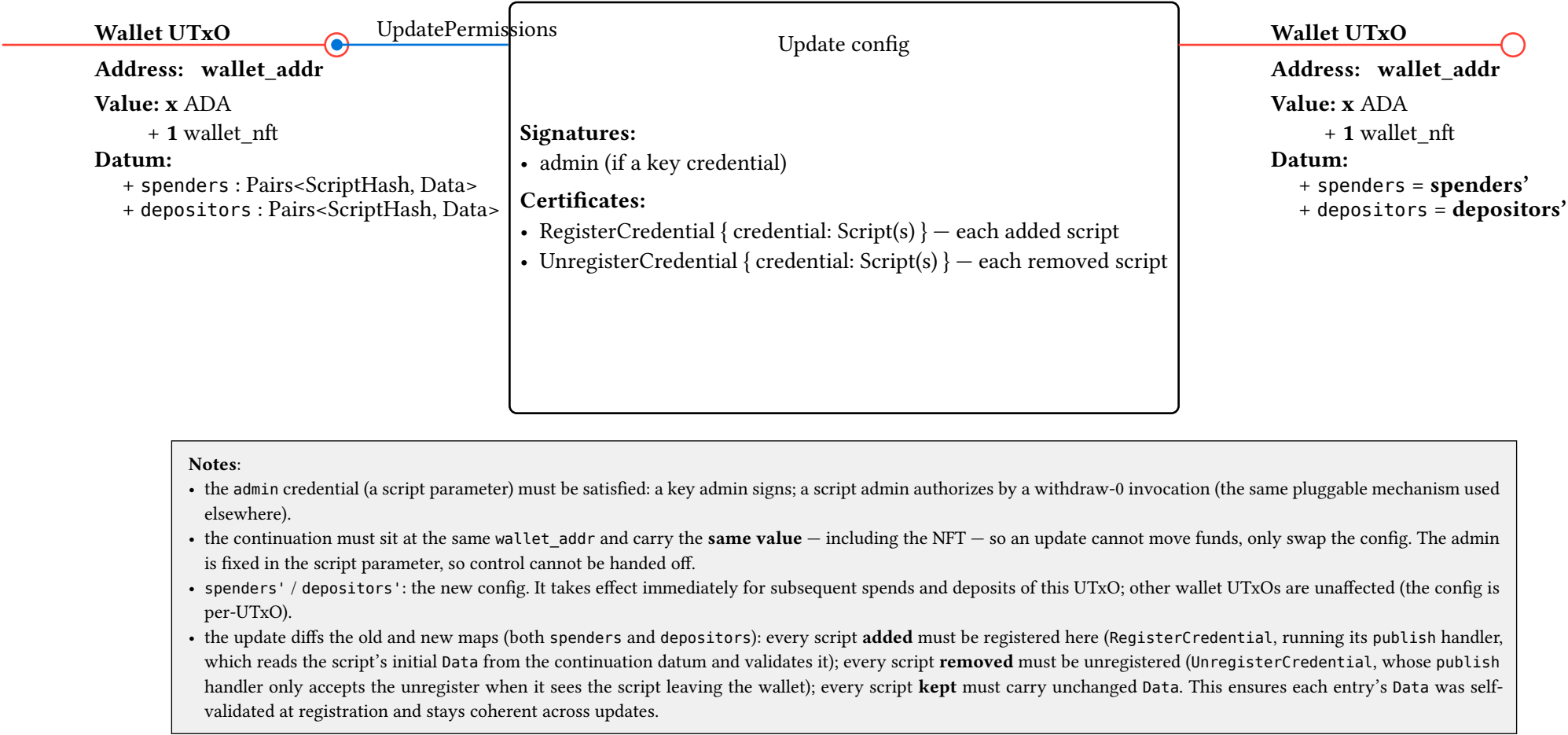
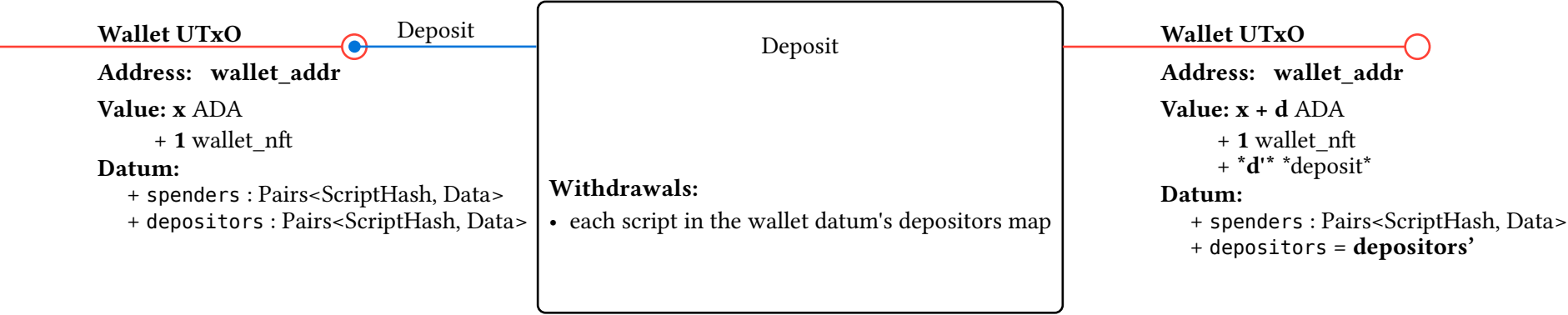


Figure 3: Update config (per-UTxO)

4 - Deposit (depositors validate added funds)

The wallet’s depositors scripts validate funds added to the wallet UTxO — the UTxO is spent and recreated at the same address with the NFT, spenders and the depositors key set preserved, and no fund removed. Every deposit script must run and approve; with none configured, a deposit is open. A deposit cannot spend funds or change the permission set.

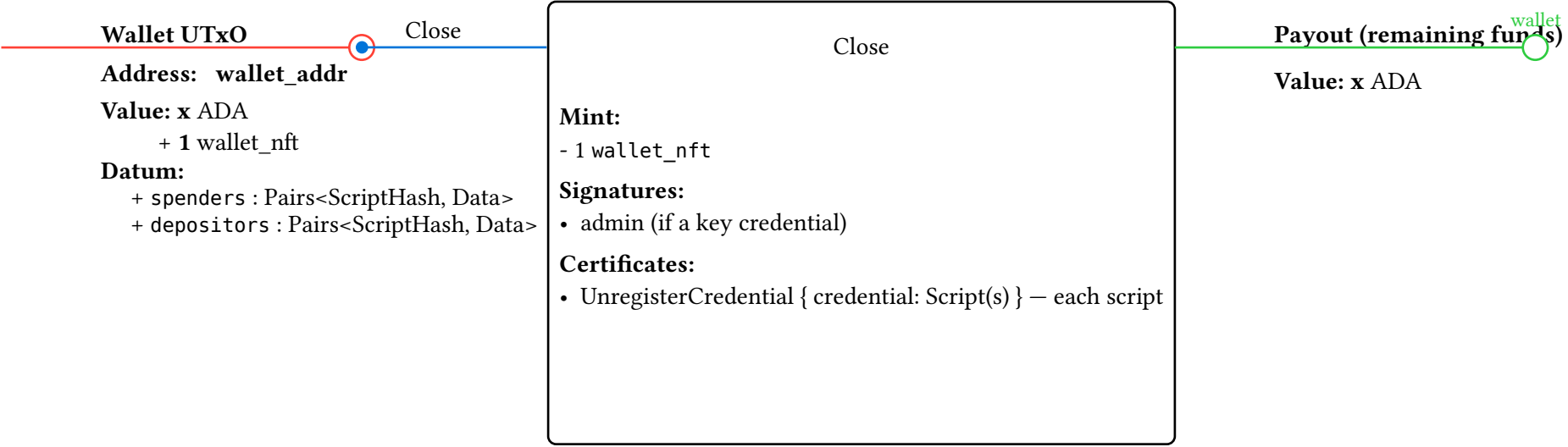


- Notes:**
- every script in the wallet UTxO’s depositors datum field is a withdraw-0 staking script that must run and approve in this transaction — e.g. asset allow/denylists, source-address checks, or a key check. With an empty map, a deposit needs no approval.
 - the continuation must sit at the same wallet_addr and preserve spenders and the depositors key set; depositor Data may advance (depositors'), with each deposit script enforcing its own input→output transition. A deposit cannot add or remove deposit scripts, or touch the spend side.
 - the continuation’s value must be a superset of the input’s: every asset (including lovelace and the NFT) is present in at least the same quantity, so a deposit can only add funds, never spend or remove them.
 - the spenders scripts do not run on a deposit: there is no outflow to restrict.

Figure 4: Deposit (depositors validate added funds)

5 - Close (admin burns the NFT)

The admin closes the wallet: the wallet UTxO is spent, releasing whatever funds remain, and the identifying NFT is burned. The Close spend redeemer requires the admin, the burn, and that every delegated script (spender and deposit) is unregistered so its stake deposit is refunded; the Burn mint redeemer permits the burn.



- Notes:**
- the admin credential (a script parameter) must be satisfied: a key admin signs; a script admin authorizes by a withdraw-0 invocation.
 - the wallet NFT — 1 wallet_nft under wallet_hash — must be burned in this transaction (-1), destroying the wallet's identity. The burn runs the mint endpoint with a Burn redeemer, which only checks the burn quantity; authorization is enforced by the Close spend redeemer.
 - closing spends the wallet UTxO and releases any remaining funds. There is no continuation: once the NFT is gone, the wallet can no longer be spent.
 - every script in the closing wallet's spenders and depositors maps must be unregistered here (UnregisterCredential), refunding its stake deposit. Each script's publish handler accepts this because it sees itself in the wallet input and no wallet output.

Figure 5: Close (admin burns the NFT)

6 - Example: stateful spending window (withdrawal script)

A stateful delegated withdrawal script that limits the wallet to at most one Spend per window. Unlike `spending_limit` (whose bound is fixed at compile time), this script carries mutable state — the last spend time — in the wallet datum’s `spenders` map, and enforces its own state transition on every spend.

The script is parameterized by `wallet` and `window` (milliseconds), and its state is a single timestamp: `SpendingWindowState { last_spend: Int }` (POSIX ms; 0 means never spent).

- **Publishing** (`RegisterCredential`): the publish handler reads its initial `Data` from the wallet output datum and requires `last_spend == 0` — an unspent wallet. Unregistering is only accepted while leaving the wallet (the same input/output presence check as `spending_limit`).
- **Withdraw** (runs on every `Spend`): the handler reads its current state from the wallet **input** datum and the next state from the wallet **output** datum, takes `now` from the validity range’s lower bound, and requires that the cooldown elapsed (`now - last_spend >= window`, or it is the first spend) and that the output state records exactly this spend (`next.last_spend == now`).
- Because the wallet’s `Spend` branch doesn’t constrain the continuation datum, the script enforces its own transition end-to-end: the builder must produce a continuation carrying the updated state, or the withdrawal fails and the transaction is rejected.

This is the template for any stateful restriction (spending budgets, time locks, allow-lists that mutate, ...): the state lives in `spenders[own_hash]`, the publish handler validates the initial value, and the `withdraw` handler reads the input state, validates the transition, and checks the output state.